

UNIT - 1

NUMBER SYSTEMS

1) BINARY NUMBER SYSTEM

↳ Base - 2
(0-1)

2) DECIMAL NUMBER SYSTEM

↳ Base - 10
(0-9)

3) OCTAL NUMBER SYSTEM

↳ Base - 8
(0-7)

4) Hexa-Decimal Number System

↳ Base - 16
(0-15)

0	7	14 (E)
1	8	15 (F)
2	9	
3	10 (A)	
4	11 (B)	
5	12 (C)	
6	13 (D)	

◦ FUNDAMENTAL DATA TYPES

- bit $\rightarrow$ 0 (OFF) & 1 (ON)
- nibble $\rightarrow$ 4 bits
- byte $\rightarrow$ 8 bits
- word $\rightarrow$ 16 bits
- double word $\rightarrow$ 32 bits

Decimal $\leftrightarrow$ Binary Tabular Representation.

Decimal	Binary (4 bit)	Decimal	Binary (4 bit)
0	0 0 0 0	9	1 0 0 1
1	0 0 0 1	10	1 0 1 0
2	0 0 1 0	11	1 0 1 1
3	0 0 1 1	12	1 1 0 0
4	0 1 0 0	13	1 1 0 1
5	0 1 0 1	14	1 1 1 0
6	0 1 1 0	15	1 1 1 1
7	0 1 1 1		
8	1 0 0 0		

CONVERSIONS

* BINARY $\xrightarrow{\text{to}}$ DECIMAL

for ex : $(1101.101)_2 \rightarrow (?.)_{10}$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 1 & . & 1 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow & \downarrow \\ 2^3 & 2^2 & 2^1 & 2^0 & & 2^{-1} & 2^{-2} & 2^{-3} \end{array}$$

$$\Rightarrow (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$$

$$\Rightarrow (8 + 4 + 0 + 1) \cdot \left(\frac{1}{2} + 0 + \frac{1}{8} \right)$$

$$\Rightarrow 13 \cdot \left(\frac{5}{8} \right) \quad \left(\frac{5}{8} = 0.625 \right)$$

$$\Rightarrow (13.625)_{10}$$

* OCTAL $\xrightarrow{\text{to}}$ DECIMAL

for ex : $(457)_8 \longrightarrow (?)_{10}$

$$\begin{array}{ccc} 4 & 5 & 7 \\ \downarrow & \downarrow & \downarrow \\ 8^2 & 8^1 & 8^0 \end{array}$$

$$\Rightarrow (4 \times 8^2) + (5 \times 8^1) + (7 \times 8^0)$$

$$\Rightarrow 256 + 40 + 7$$

$$\Rightarrow (303)_{10}$$

* HEXA-DECIMAL $\xrightarrow{\text{to}}$ DECIMAL

for ex : $(1DF)_{16} \rightarrow (?)_{10}$

$$\begin{array}{ccc} 1 & D & F \\ \downarrow & \downarrow & \downarrow \\ 16^2 & 16^1 & 16^0 \end{array}$$

$$\Rightarrow (1 \times 16^2) + (D \times 16^1) + (F \times 16^0)$$

$$\Rightarrow 16^2 + 13 \times 16 + 15 \times 1$$

$$\Rightarrow 256 + 208 + 15$$

$$\Rightarrow (479)_{10}$$

* DECIMAL $\xrightarrow{\text{to}}$ BINARY

for ex : (i) $(37)_{10} \longrightarrow (?)_2$

2	37	1	↑
2	18	0	
2	9	1	
2	4	0	
2	2	0	
	1		↑

$$\Rightarrow (100101)_2$$

(ii) $(0.8125)_{10} \rightarrow (?)_2$

$0.8125 \times 2 = 1.625$ 1

$0.6250 \times 2 = 1.25$ 1

$0.2500 \times 2 = 0.5$ 0

$0.5000 \times 2 = 1.0$ 1

$= (0.1101)_2$

(iii) $(0.95)_{10} \rightarrow (?)_2$

$0.95 \times 2 = 1.9$ 1

$0.90 \times 2 = 1.8$ 1

$0.80 \times 2 = 1.6$ 1

$0.60 \times 2 = 1.2$ 1

$0.20 \times 2 = 0.4$ 0

$0.40 \times 2 = 0.8$ 0

$0.80 \times 2 = 1.6$ 1

$= (0.1111001)_2$

(iv) $(24.6)_{10} \rightarrow (?)_2$

2 | 24 | 0

2 | 12 | 0

2 | 6 | 0

2 | 3 | 1

1

$0.6 \times 2 = 1.2$ 1

$0.2 \times 2 = 0.4$ 0

$0.4 \times 2 = 0.8$ 0

$0.8 \times 2 = 1.6$ 1

$0.6 \times 2 = 1.2$ 1

$0.2 \times 2 = 0.4$ 0

$= 11000$

Answer $\rightarrow (11000.100110)_2$

* DECIMAL $\xrightarrow{\text{to}}$ OCTAL

for ex : $(35.45)_{10} \longrightarrow (?)_8$

8	35	3	↑
	4		

$0.45 \times 8 = 3.6 \quad 3$

$0.60 \times 8 = 4.8 \quad 4$

$0.8 \times 8 = 6.4 \quad 6$

$0.4 \times 8 = 3.2 \quad 3$

$0.2 \times 8 = 1.6 \quad 1$

Answer $\rightarrow (43.34631)_8$

* DECIMAL $\xrightarrow{\text{to}}$ HEXADECIMAL

for ex : $(22.64)_{10} \longrightarrow (?)_{16}$

16	22	6	↑
	1		

$0.64 \times 16 = 10.24 \quad A$

$0.24 \times 16 = 3.84 \quad 3$

$0.84 \times 16 = 13.44 \quad D$

$0.44 \times 16 = 7.04 \quad 7$

Answer $\rightarrow (16.A3D7)_{16}$

* OCTAL $\xrightarrow{\text{to}}$ BINARY

for ex : $(3764)_8 \longrightarrow (?)_2$

(first to decimal)

3 7 6 4
↓ ↓ ↓ ↓
 $8^3 \ 8^2 \ 8^1 \ 8^0$

$$\rightarrow (3 \times 8^3) + (7 \times 8^2) + (6 \times 8^1) + (4 \times 1)$$

$$\rightarrow 1536 + 448 + 48 + 4$$

$$\rightarrow (2036)_{10}$$

(Then decimal to Binary)

2	2036	0
2	1018	0
2	509	1
2	254	0
2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
1	1	

Answer $\rightarrow (11111110100)_2$

* BINARY $\xrightarrow{\text{to}}$ OCTAL & OCTAL $\xrightarrow{\text{to}}$ BINARY
(Alternate Way)

$\rightarrow$ for converting Binary to Octal, divide the Binary number in 3 bits and then convert those sets of 3 bits to decimal no. The final no. we get is an Octal No.

$\rightarrow$ for converting octal back to binary, do the reverse.

for ex : $(111101100)_2 \rightarrow (?)_8$

(direction of dividing : from Right towards left.
3 bits)

$\begin{array}{ccc} \underline{111} & \underline{101} & \underline{100} \\ 7 & 5 & 4 \end{array}$

$\Rightarrow (754)_8$

(ii) $(1100100010)_2 \longrightarrow (?)_8$

$\begin{array}{cccc} \underline{1100} & \underline{1000} & \underline{10} & \underline{10} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 001 & 100 & 100 & 010 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 4 & 4 & 2 \end{array}$

$\Rightarrow (1442)_8$

(iii) $(7036)_8 \longrightarrow (?)_2$

$\begin{array}{cccc} 7 & 0 & 3 & 6 \\ \swarrow & \downarrow & \downarrow & \searrow \\ 111 & 000 & 011 & 110 \end{array}$

$\Rightarrow (111000011110)_2$

* BINARY $\rightarrow$ HEXA-DECIMAL $\neq$ HEXADECIMAL $\rightarrow$ BINARY

$\rightarrow$ for Converting Binary to Hexadecimal, divide the Binary no. in 4 bits And then convert those sets of 4 bits to decimal no. The final no. obtained is a Hexadecimal Number.

$\rightarrow$ for Converting Hexadecimal back to Binary, just do the reverse.

for ex : (i) $(00111111100)_2 \rightarrow (?)_{16}$

$$\begin{array}{cccc} 0011 & 1111 & 1100 \\ \downarrow & \downarrow & \downarrow \\ 3 & 15 & 12 \\ (F) & (F) & (C) \end{array}$$

(direction of $\rightarrow$ Right towards left
dividing)

$\Rightarrow (3FC)_{16}$

(ii) $(DB9B)_{16} \rightarrow (?)_2$

$$\begin{array}{cccc} D & 8 & 9 & B \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 13 & 8 & 9 & 11 \\ \swarrow & \downarrow & \searrow & \searrow \\ 1101 & 1000 & 1001 & 1011 \end{array}$$

$\Rightarrow (1101100010011011)_2$

BINARY ADDITION

	Sum	Carry
$0+0 =$	0	0
$0+1 =$	1	0
$1+0 =$	1	0
$1+1 =$	0	1

for ex :

$$\begin{array}{r}
 \textcircled{1} \textcircled{1} \textcircled{1} \\
 1001.011 \\
 + 0100.110 \\
 \hline
 1110.001
 \end{array}$$

BINARY SUBTRACTION

	Diff.	Borrow
$0-0 =$	0	0
$0-1 =$	1	1
$1-0 =$	1	0
$1-1 =$	0	0

for ex :

(i)

$$\begin{array}{r}
 10010 \\
 - 100111 \\
 \hline
 001011
 \end{array}$$

(ii)

$$\begin{array}{r}
 1001001 \\
 - 0101010 \\
 \hline
 01000111
 \end{array}$$

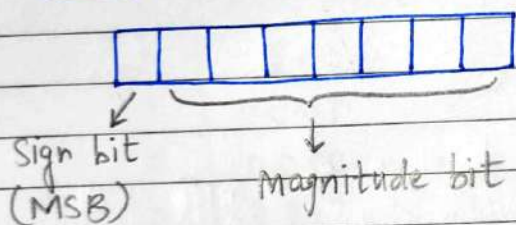
$$\begin{array}{r}
 \text{(iii)} \quad 100110 \\
 - 011101 \\
 \hline
 001001
 \end{array}$$

SIGNED BINARY NUMBERS

The 3 types of representation are as follows :-

1. Sign Magnitude form.
2. 1's Complement form.
3. 2's Complement form.

8-bit



MSB (Most Significant bit)
→ Leftmost bit.
LSB (Least Significant bit)
→ Rightmost bit.

- If the value of sign bit is 0, then it will represent +ve Sign.
- While, if the value of sign bit is 1, then it will represent -ve Sign.

1's Complement

→ Invert all bits of the given binary no.
i.e.

0 → 1

1 → 0

2's Complement

It is obtained by adding 1 to the LSB of the no.
for ex : Number → 101101

1's complement → 010010

2's complement → 010010
+1
010011

1) Signed Magnitude form

+25 (MSB)

0	0	0	1	1	0	0	1
---	---	---	---	---	---	---	---

↓
+ve sign bit

-25

1	0	0	1	1	0	0	1
---	---	---	---	---	---	---	---

↓
-ve sign bit

2) 1's Complement form

+25

0	0	0	1	1	0	0	1
---	---	---	---	---	---	---	---

(+ve)

-25

1	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---

(-ve)

3) 2's Complement form

+25

0	0	0	1	1	0	0	1
---	---	---	---	---	---	---	---

-25

1	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---

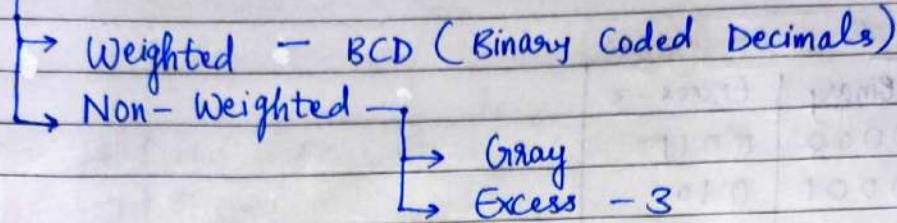
(1's form) + 1

-25

1	1	1	0	0	1	1	1
---	---	---	---	---	---	---	---

(2's form) ↓
-ve sign bit

BINARY CODES



WEIGHTED

o BCD Code

→ Each decimal digit (0-9) is represented by its equivalent 4-bit binary code.

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

for ex : (i) 43

- Binary eq. = 101011
- BCD = 0100 0011
(4) (3)

(ii) 625.81

- BCD = 0110 0010 0101 . 0011 0001
(6) (2) (5) (3) (1)

NON - WEIGHTED

• Excess - 3

Decimal	Binary	⁽⁺³⁾ Excess - 3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

for ex : (i) 425.32

• BCD = 0100 0010 0101 . 0011 0010

• Excess-3 = 0111 0101 1000 . 0110 0101

• Gray Codes

→ It shows only a single bit change from one codeword to the next in sequence (and in such a way, that the first half of the codes is the reflection of the 2nd half.)

→ These codes are also known as Reflexive Code / Unit Distance Code / Cyclic Code.

* Gray Codes (in 2-bit)

0	00
1	01
3	11
2	10

(mirror)

($2_{10} / 10_2$ didn't come before/prior because if it does, change in 2 bits would have been observed.)

* Gray Codes (in 3-bit)

0	000
1	001
3	011
2	010
6	110
7	111
5	101
4	100

(mirror)

→ Similarly, we can write Gray Codes (in 4-bit)

0	0000
1	0001
3	0011
2	0010
6	0110
7	0111
5	0101
4	0100
12	1100
13	1101
15	1111
14	1110
10	1010
11	1011
9	1001
8	1000

(mirror)

CODES CONVERSION

* BINARY TO GRAY

→ Write down the MSB as it is, then perform XOR $\oplus$ operation on the consecutive bit simultaneously and write the result.

for ex : (i) 101101
 $\oplus \oplus \oplus \oplus$
 101101
 ↓
 111011

(ii) 1001101101

1001101101
 ↓
 1101011011

* GRAY to BINARY

for ex : 10010

10010
 $\downarrow \uparrow \uparrow \uparrow$
 11100

Answer → 11100

BINARY SUBTRACTION USING COMPLEMENT METHOD

r 's Complement Method

($r \rightarrow$ value of Base)

$$M - N$$

1. find r 's complement of N .
2. Add it to M .
3. If carry is generated, discard it. (It means result is true & true ($M > N$).)
4. If No carry is generated, to obtain the true result, find r 's complement of sum & place a -ve sign before it. It means $M < N$.

Ques.) (i) $(1011)_2 - (0100)_2$ using 2's complement method.
(M) (N)

Ans. (i) $r = 2$

1. 2's complement of 0100 (N)

$$\begin{array}{r} 0100 \\ \downarrow \oplus \\ 1011 \\ + 1 \\ \hline 1100 \end{array}$$

$$\begin{array}{r} \oplus \\ 1100 \\ + 1011 \\ \hline \end{array}$$

Carry $\leftarrow$ (discarded it)

Ans. $\rightarrow (0111)_2$

$$(ii) \quad (0100)_2 - (1001)_2$$

(M) (N)

Ans (ii)

$$r=2$$

$$1) \quad 1001$$

↓

$$0110$$

$$+ 1$$

$$\hline 0111$$

⊕

$$2) \quad 0111$$

$$+ 0100$$

$$\hline 1011$$

(No carry)

co₂

$$1011$$

↓

2's complement

$$0100$$

$$+ \quad 1$$

$$\hline 0101$$

Ans. $\rightarrow (0101)_2 \rightarrow (-5)_{10}$

(r-1)'s complement method

$$M - N$$

1. Find $(r-1)$'s complement of N.
2. Add it to M.
3. If carry is generated discard it from that position and add it to the LSB of sum (It means $M > N$).

(Least Significant Bit
(Right-most Bit))

4. If No carry is generated, to obtain the true result, find $(r-1)$'s complement of sum & place -ve sign before it. (It means $M < N$).

Ques.) (i) $(1011)_2 - (0100)_2$ using 1's complement method.
 $(M) \quad (N)$

Ans.(i)

$$r = 2$$

$$r-1 = 1$$

So,

$$\begin{array}{r} 0100 \\ \textcircled{1} 010 \\ + 1011 \\ \hline \textcircled{1}0110 \end{array} \quad (1's \text{ complement})$$

$$\begin{array}{r} 0110 \\ + 1 \\ \hline 0111 \end{array}$$

Ans. $\rightarrow (0111)_2 \rightarrow (7)_{10}$

(ii) $(0100)_2 - (1001)_2$ using 1's complement method.
 $(M) \quad (N)$

Ans.(ii)

$$\begin{array}{r} 1001 \\ \downarrow (1's \text{ complement}) \\ \textcircled{1}0110 \\ + 0100 \\ \hline 1010 \end{array}$$

$$\begin{array}{r} 0101 \\ - 0101 \end{array} \quad (1's \text{ complement of the sum})$$

Sq, Answer $\rightarrow (0101)_2 \rightarrow (-5)_{10}$

o How to find 7's, 8's, 9's, 10's, 15's, 16's Complement

• 7's and 8's complement

for 7's complement, subtract each bit of the no. from 7.

And for 8's complement, add 1 to the 7's complement.

(This method applies for 9's, 10's & 15's, 16's complement as well.)

for ex : (i) $(1635)_8$

$$\begin{array}{r} 7 \ 7 \ 7 \ 7 \\ - 1 \ 6 \ 3 \ 5 \\ \hline 6 \ 1 \ 4 \ 2 \end{array}$$

$$\begin{array}{r} 6 \ 1 \ 4 \ 2 \\ + 1 \\ \hline 6 \ 1 \ 4 \ 3 \end{array}$$

(ii) $(2ACD)_{16}$

$$\begin{array}{r} F \ F \ F \ F \\ - 2 \ A \ C \ D \\ \hline D \ 5 \ 3 \ 2 \end{array}$$

(Subtracting each bit from 15 (F))

16's complement $\rightarrow$

$$\begin{array}{r}
 D\ 5\ 3\ 2 \\
 +\ 1 \\
 \hline
 D\ 5\ 3\ 3
 \end{array}$$

Homework Questions

- Ques. 1 (a) $(51)_{10} - (22)_{10}$ using 10's complement method.
 (b) $(4)_{10} - (7)_{10}$ using 9's complement method.
 (c) $(52)_{10} - (89)_{10}$ using 9's complement method.
 (d) $(2ADC)_{16}$ F's complement.
 (e) $(261)_8$ 7's complement.
 (f) $X = (1010100)_2$, then find (i) $X - Y$ (using 2's complement).
 $Y = (1000011)_2$ (ii) $Y - X$ (using 2's complement).

Ans. 1 (a) $(51)_{10} - (22)_{10}$ (Using 10's comp.)
 (M) (N)

$$\begin{array}{r}
 99 \\
 - 22 \\
 \hline
 77 \quad \Rightarrow \quad \begin{array}{r} 77 \\ + 1 \\ \hline 78 \end{array} \quad (10's \text{ complement})
 \end{array}$$

So,

$$\begin{array}{r}
 78 \\
 + 51 \\
 \hline
 132
 \end{array}$$

(Discard it)

Answer $\rightarrow (32)_{10}$

(b) $(4)_{10} - (7)_{10}$ (using 9's comp.)

$$\begin{array}{r} 9 \\ - 7 \\ \hline 2 \end{array}$$

So,

$$\begin{array}{r} 2 \\ + 4 \\ \hline 6 \end{array}$$

(No carry)

$$\begin{array}{r} 9 \\ - 6 \\ \hline 3 \end{array}$$

So,

Answer $\rightarrow (-3)_{10}$

(c) $(52)_{10} - (89)_{10}$ (Using 9's comp.)

$$\begin{array}{r} 99 \\ - 89 \\ \hline 10 \end{array}$$

So,

$$\begin{array}{r} 10 \\ + 52 \\ \hline 62 \end{array}$$

(No carry)

$$\begin{array}{r} 99 \\ - 62 \\ \hline 37 \end{array}$$

So,

Answer $\rightarrow (-37)_{10}$

(d) $(2ADC)_{16}$ F's (15's) Complement.

$$\begin{array}{r} F F F F \\ - 2 A D C \\ \hline D 5 2 3 \end{array}$$

Answer $\rightarrow (D523)_{16}$

(e) $(261)_8$ 7's complement

$$\begin{array}{r} 777 \\ - 261 \\ \hline 516 \end{array}$$

Answer $\rightarrow (516)_8$

(f) $X = (1010100)_2$
 $Y = (1000011)_2$

$X - Y$ (using 2's complement)
 $Y - X$

$X - Y$

$$\begin{array}{r} 1000011 \\ \downarrow \text{(1's comp.)} \\ 0111100 \\ + 1 \\ \hline 0111101 \text{ (2's comp.)} \end{array}$$

$$\begin{array}{r} 0111101 \\ + 1010100 \\ \hline 10010001 \end{array}$$

(discard it)

Answer $\rightarrow X - Y = 0010001$
 $(17)_{10}$

$Y - X$

$$\begin{array}{r} 1010100 \\ \downarrow \\ \text{(1's comp.) } 0101011 \\ + 1 \\ \hline \text{(2's comp.) } 0101100 \end{array}$$

$$\begin{array}{r} 0101100 \\ + 1000011 \\ \hline 1101111 \text{ (No carry)} \\ \downarrow \\ 0010000 \\ + 1 \\ \hline 0010001 \end{array}$$

Answer $\rightarrow (0010001)_2$
 OR $(-17)_{10}$

BCD ADDITION

Sum ≤ 9
& Carry 0

→ Result is true
no correction is
required.

Sum ≤ 9
& Carry 1.

→ To obtain True
result add '6' to
the sum.

Sum > 9
& carry 0

→ Add 6 to the
Sum.

For ex : (i) $(2)_{10} + (6)_{10}$

$$\begin{array}{r} 0010 \\ + 0110 \\ \hline 1000 \end{array}$$

Answer → $(8)_{10}$

(ii) $(7)_{10} \& (6)_{10}$

$$\begin{array}{r} 0111 \\ + 0110 \\ \hline 1101 \end{array} \leftarrow (\text{BxD})$$

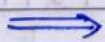
say 0 0

$$\begin{array}{r} 1101 \\ + 0110 \\ \hline 10011 \end{array}$$

Answer → 0001 0011
 $(13)_{10}$

(iii) $(9)_{10} + (8)_{10}$

$$\begin{array}{r} 1001 \\ + 1000 \\ \hline 10001 \end{array}$$



$$\begin{array}{r} 0001\ 0001 \\ + 0000\ 0110 \\ \hline 0001\ 0111 \end{array}$$

Answer → 0001 0111
 $(17)_{10}$

(iv) $(57)_{10} + (26)_{10}$

$$\begin{array}{r}
 01010111 \\
 + 00100110 \\
 \hline
 01111101
 \end{array}$$

so,

$$\begin{array}{r}
 01111101 \\
 + 00000110 \\
 \hline
 10000011
 \end{array}$$

Answer $\rightarrow$ 1000 0011
 $(83)_{10}$

(v) $(83)_{10} + (34)_{10}$

$$\begin{array}{r}
 10000011 \\
 + 00110100 \\
 \hline
 10110111
 \end{array}$$

so,

$$\begin{array}{r}
 10110111 \\
 + 01100000 \\
 \hline
 100010001011
 \end{array}$$

so, Answer $\rightarrow$ 0001 0001 0111
 $(117)_{10}$

(vi) $(569)_{10} + (687)_{10}$

$$\begin{array}{r}
 010101101001 \\
 + 011010000111 \\
 \hline
 10111110000
 \end{array}$$

so,

$$\begin{array}{r}
 10111110000 \\
 + 011001100110 \\
 \hline
 10001001010110
 \end{array}$$

Answer $\rightarrow$ 0001 0010 0101 0110
 $(1256)_{10}$

BOOLEAN ALGEBRA

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + \bar{A} = 1$$

$$A + A = A$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A \cdot \bar{A} = 0$$

$$A \cdot A = A$$

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

(Commutative Law).

$$A + (B + C) = (A + B) + C$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

(Associative Law)

$$A(B + C) = AB + AC$$

$$A + BC = (A + B)(A + C)$$

(Distributive Law)

Proof for : $A + BC = (A + B)(A + C)$

so, R.H.S $\Rightarrow (A + B) \cdot (A + C)$

$$\Rightarrow A \cdot A + A \cdot C + B \cdot A + B \cdot C$$

$$\Rightarrow A + A \cdot C + BA + B \cdot C$$

$$\Rightarrow A(1 + C) + BA + B \cdot C$$

$$\Rightarrow A(1) + BA + B \cdot C$$

$$\Rightarrow A(1 + B) + B \cdot C$$

$$\Rightarrow A + B \cdot C$$

Hence Proved

Ques.) Simplify the following Boolean Expression:

(i) $A\bar{B}C + \bar{A}BC + ABC$

Ans(i) $(A\bar{B})C + \bar{A}BC + ABC$

$$\Rightarrow AC(\bar{B} + B) + \bar{A}BC$$

$$\Rightarrow AC(1) + \bar{A}BC$$

$$\Rightarrow C(A + \bar{A}B)$$

$$\Rightarrow C((A + \bar{A})(A + B))$$

$$\Rightarrow C(A + B)$$

(ii) $Y = (AB + C)(AB + D)$

Ans(ii) $Y = ABAB + ABD + ABC + CD$
 $= (A \cdot A)(B \cdot B) + ABD + ABC + CD$

$$\Rightarrow AB + ABC + ABD + CD$$

$$= AB[(1 + C) + D] + CD$$

$$= AB[1 + D] + CD$$

$$= AB + CD$$

(iii) $Y = \overline{AB} + \overline{A} + AB$

De Morgan's Theorem :

$$\left. \begin{aligned} \overline{X+Y} &= \overline{X} \cdot \overline{Y} \\ \overline{XY} &= \overline{X} + \overline{Y} \end{aligned} \right\}$$

Ans (iii) $\overline{AB} \cdot \overline{A} \cdot \overline{AB}$

$$\rightarrow \overline{AB} \cdot \overline{A} \cdot (\overline{A} + \overline{B})$$

$$\Rightarrow (\overline{A} \cdot \overline{A}) \overline{B} \cdot (\overline{A} + \overline{B})$$

$$\Rightarrow \overline{A} \overline{B} (\overline{A} + \overline{B})$$

$$\Rightarrow \overline{A} \overline{A} \overline{B} + \overline{A} \overline{B} \overline{B}$$

$$\Rightarrow 0 + 0$$

$$\Rightarrow 0$$

(iv) If $X=1$ in the logic eqⁿ. :-

$$\{X+Z [\overline{Y} + (\overline{Z} + X\overline{Y})]\} \{ \overline{X} + \overline{Z}(X+Y) \} = 1$$

then, choose :

(a) $Y=Z$

(c) $Y = \overline{Z}$

(b) $Z=1$

(d) $Z=0$

Ans: $\{1+z[\bar{Y}+(\bar{Z}+1\cdot\bar{Y})]\}\{0+\bar{Z}(1+Y)\} = 1$

$\rightarrow \{1+z[\bar{Y}+\bar{Z}+\bar{Y}]\}\{0+\bar{Z}(1)\} = 1$

$\rightarrow \{1+z[\bar{Z}+\bar{Y}]\}\{\bar{Z}\} = 1$

$\rightarrow (1+z\bar{Z}+z\bar{Y})\bar{Z} = 1$

$\rightarrow (1+z\bar{Y})\bar{Z} = 1$

$\rightarrow \bar{Z} + z\bar{Z}\bar{Y} = 1$

$\Rightarrow \bar{Z} = 1$ i.e.

$Z = 0$

Answer $\rightarrow$ (d) $Z=0$

□ SOP (Sum of Product) & POS (Product of Sum)

Sum term $\rightarrow A+B$

Product term $\rightarrow A\cdot B$

SOP (Sum of Product)

for ex $\rightarrow AB + BC + ABC$

Standard / Canonical SOP form: A standard SOP is one in which all the product terms (product of literals) are min terms.

(min Term is a product term containing all the variables of the function either in complemented or in uncomplemented form.)

Literals $\rightarrow$ combination of variable & its complement

Min Terms (m)

$\bar{A} \bar{B} \bar{C}$	0 0 0	m_0
$\bar{A} \bar{B} C$	0 0 1	m_1
$\bar{A} B \bar{C}$	0 1 0	m_2
$\bar{A} B C$	0 1 1	m_3
$A \bar{B} \bar{C}$	1 0 0	m_4
$A \bar{B} C$	1 0 1	m_5
$A B \bar{C}$	1 1 0	m_6
$A B C$	1 1 1	m_7

• POS (Product of Sum)

for ex $\rightarrow (A+B+C) \cdot (A+B)$

Standard / Canonical POS form : In which all the sum terms are max. terms.

max term $\rightarrow$ is a sum term which contains all the variables of the function either in complemented or uncomplemented form.

for ex : $A\bar{B}C + AB$ (Not in Std. form)

$$\Rightarrow A\bar{B}C + AB(C + \bar{C})$$

$$\Rightarrow A\bar{B}C + ABC + AB\bar{C} \quad \left(\begin{array}{l} \text{Converted to} \\ \text{Std. form} \end{array} \right)$$

Max Terms (M)

$A+B+C$	000	M_0
$A+B+\bar{C}$	001	M_1
$A+\bar{B}+C$	010	M_2
$A+\bar{B}+\bar{C}$	011	M_3
$\bar{A}+B+C$	100	M_4
$\bar{A}+B+\bar{C}$	101	M_5
$\bar{A}+\bar{B}+C$	110	M_6
$\bar{A}+\bar{B}+\bar{C}$	111	M_7

Ques.) (i) Convert to standard SOP form: $Y = \bar{A}\bar{B}\bar{C} + ABC + B\bar{C}$.

Ans (i) $Y = \bar{A}\bar{B}\bar{C} + ABC + B\bar{C}(A+\bar{A})$ ($A+\bar{A} = 1$)

$$\Rightarrow Y = \bar{A}\bar{B}\bar{C} + ABC + AB\bar{C} + \bar{A}B\bar{C}$$

(ii) Convert to standard POS form: $Y = A(B+C)(C+\bar{A})$

Ans (ii) $\Rightarrow Y = (A+B\bar{B}+C\bar{C})(B+C)(C+\bar{A})$

$$\Rightarrow Y = (A+B\bar{B}+C\bar{C})(A\bar{A}+B+C)(\bar{A}+B\bar{B}+C)$$

(iii) Convert to standard POS form: $Y = (A+B)(\bar{A}+C)(B+C+\bar{A})$

Ans (iii) $Y = (A+B+C\bar{C})(\bar{A}+B\bar{B}+C)(B+C+\bar{A})$

$$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)(B+C+\bar{A})$$

(m_7)

(m_6)

(m_3)

(m_1)

(m_0)

Max Terms (M)

$A+B+C$	000	M_0
$A+B+\bar{C}$	001	M_1
$A+\bar{B}+C$	010	M_2
$A+\bar{B}+\bar{C}$	011	M_3
$\bar{A}+B+C$	100	M_4
$\bar{A}+B+\bar{C}$	101	M_5
$\bar{A}+\bar{B}+C$	110	M_6
$\bar{A}+\bar{B}+\bar{C}$	111	M_7

Ques. (i) Convert to standard SOP form: $Y = \bar{A}\bar{B}\bar{C} + ABC + B\bar{C}$

Ans. (i) $Y = \bar{A}\bar{B}\bar{C} + ABC + B\bar{C}(A + \bar{A}) \quad (A + \bar{A} = 1)$

$\Rightarrow Y = \bar{A}\bar{B}\bar{C} + ABC + AB\bar{C} + \bar{A}B\bar{C}$

(ii) Convert to standard POS form: $Y = A(B+C)(C+\bar{A})$

Ans. (ii) $\Rightarrow Y = (A + B\bar{B} + C\bar{C})(B+C)(C+\bar{A})$

$\Rightarrow Y = (A + B\bar{B} + C\bar{C})(A\bar{A} + B + C)(\bar{A} + B\bar{B} + C)$

(iii) Convert to standard POS form: $Y = (A+B)(\bar{A}+C)(B+C+\bar{A})$

Ans. (iii) $Y = (A+B+C\bar{C})(\bar{A}+B\bar{B}+C)(B+C+\bar{A})$

$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)(B+C+\bar{A})$

(m_7) (m_6) (m_3) (m_1) (m_0)

K-MAP (KARNAUGH MAP)

→ K-maps are used to simplify Boolean Expression.

SOP form :

1) K-map of 2-Variables.

A \ B	0	1
0	m_0	m_1
1	m_2	m_3

Total Blocks = 2^n
($n \rightarrow$ no. of variables)

$$= 2^2 = 4$$

Lone	→	Single 1's		(2 lones)
Pair	→	2 1's (always adjacent)		
Quad	→	4 1's		
Octet	→	8 1's		

Precedence :- Octet > Quad > Pair > lone.

Note: Always make the min. possible pair, quad, etc.

2) K-map of 3-Variables.

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Total Blocks = 2^3
= 8

3) K-map of 4-variables

AB \ CD	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$$\text{Total Blocks} = 2^4 = 16.$$

for ex : (i)

A \ B	0	1
0		1
1	1	

(2 ones)

$$Y = \bar{A}B + A\bar{B}$$

(ii)

A \ B	0	1
0	1	1
1	1	

(2 pairs)

$$Y = \bar{A} + \bar{B}$$

(iii)

A \ BC	00	01	11	10
0		1	1	
1		1	1	1

$$Y = C + AB$$

(1 Quad, 1 pair)

(iv)

A \ BC	00	01	11	10
0	1	1	1	1
1	1			1

$$Y = \bar{A} + \bar{C}$$

(2 Quads)

Boolean

Ques.) Write the corresponding expression for given K-maps:

(i) $Y = \sum m(0, 1, 2, 5, 13, 15)$

Ans(i):

AB \ CD	00	01	11	10
00	1	1		1
01		1		
11		1	1	
10				

$Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C}D + ABD$

(3 pairs)

(ii) $Y = \sum m(1, 5, 7, 9, 11, 13, 15)$

AB \ CD	00	01	11	10
00		1		
01		1	1	
11		1	1	
10		1	1	

$Y = \bar{C}D + BD + AD$

(3 Quads)

(iii) $Y = \sum m(1, 3, 5, 9, 11, 13)$

AB \ CD	00	01	11	10
00		1	1	
01		1		
11		1		
10		1	1	

$Y = \bar{C}D + \bar{B}D$

(2 Quads)

(iv.) $Y = \sum m(1, 2, 9, 10, 11, 14, 15)$

AB \ CD	00	01	11	10
00		1		1
01				
11			1	1
10		1	1	1

(1 Quad,
2 pairs)

$$Y = AC + \bar{B}\bar{C}D + \bar{B}C\bar{D}$$

(v.) $Y = \sum m(4, 5, 8, 9, 11, 12, 13, 15)$

AB \ CD	00	01	11	10
00				
01	1	1		
11	1	1	1	
10	1	1	1	

(3 Quads)

$$Y = B\bar{C} + A\bar{C} + AD$$

• Redundant Group → If all 1's in a group are already involved in some other groups then that group is called the Redundant Group.

And, the term that it constitutes must be discarded.

(vi.) $Y = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

AB \ CD	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

(1 Quad
(redundant)
4 pairs)

$$Y = \underbrace{BD}_{\text{discarded}} + \bar{A}\bar{C}D + AB\bar{C} + ACD + \bar{A}BC$$

So,

Answer :-

$$Y = \bar{A}\bar{C}D + AB\bar{C} + ACD + \bar{A}BC$$

(vii) $Y = \sum m(0, 1, 2, 5, 13, 15)$

(3 pair)

AB \ CD	00	01	11	10
00	1	1		1
01		1		
11		1	1	
10				

$$Y = \bar{A}\bar{B}\bar{D} + \bar{A}\bar{C}D + ABD$$

(viii) $Y = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$

$d \rightarrow$ don't care condition
(represented by X)

$\rightarrow$ We will include X in pairs, quads, octet unless all the 1's are covered. The moment all the 1's are covered and some X are still left, no need to include them in quad, pair, etc anymore

(2 quads)

AB \ CD	00	01	11	10
00	X	1	1	X
01		X	1	
11			1	
10			1	

$$Y = \bar{A}\bar{B} + CD$$

(ix) $Y = \sum m(1, 4, 8, 12, 13, 15) + d(3, 14)$

(1 quad, 3 pairs)

AB \ CD	00	01	11	10
00		1	X	
01	1			
11	X	1	1	X
10	1			

$$Y = AB + \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D}$$

(x) $Y = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5, 8, 14)$

(2 Quads)

AB \ CD	00	01	11	10
00	X	1	1	X
01		X	1	
11			1	X
10	X		1	

$Y = \bar{A}D + CD$

(xi) $Y = \sum m(0, 1, 5, 9, 13, 14, 15) + d(3, 4, 7, 10, 11)$

(1 Octet, 2 Quads)

AB \ CD	00	01	11	10
00	1	1	X	
01	X	1	X	
11		1	1	1
10		1	X	X

$Y = D + \bar{A}\bar{C} + AC$

POS FORM

(i) $Y = \prod M(0, 2, 3, 5, 7)$

A \ BC	00	01	11	10
0	0		0	0
1		0	0	

$Y = (A+C) \cdot (\bar{B}+\bar{C}) \cdot (\bar{A}+\bar{C})$

(ii)

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	0	0	0
11	1	1	1	0
10	0	0	1	0

$Y = \bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}C + AB\bar{C} + ACD$
(SOP)

$Y = (A+C+\bar{D}) \cdot (A+\bar{B}+\bar{C}) \cdot (\bar{A}+B+C) \cdot (\bar{A}+\bar{C}+D)$
(POS)

Q-M (Quine - McCluskey) Method

for instance : (i) $Y = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$

(I)	A	B	C	D		(II)	A	B	C	D
$(m_0) 0$	0	0	0	0	(Zero 1's)	(m_0, m_1)	0	0	0	—
$(m_1) 1$	0	0	0	1	(One 1's)	(m_0, m_8)	—	0	0	0
$(m_8) 8$	1	0	0	0		(m_1, m_3)	0	0	—	1
$(m_3) 3$	0	0	1	1	(Two 1's)	(m_1, m_9)	—	0	0	1
$(m_9) 9$	1	0	0	1		(m_8, m_9)	1	0	0	—
$(m_7) 7$	0	1	1	1	(Three 1's)	(m_3, m_7)	0	—	1	1
$(m_{11}) 11$	1	0	1	1		(m_3, m_{11})	—	0	1	1
$(m_{15}) 15$	1	1	1	1	(Four 1's)	(m_9, m_{11})	1	0	—	1
						(m_7, m_{15})	—	1	1	1
						(m_{11}, m_{15})	1	—	1	1

→ Make pairs of min terms in Table (II) in such a way that change in exactly single bit takes place.

(III)	A	B	C	D	Prime Implicant (PI)
(m_0, m_1, m_8, m_9)	—	0	0	—	$\bar{B}\bar{C}$
(m_0, m_8, m_1, m_9)	—	0	0	—	
(m_1, m_3, m_9, m_{11})	—	0	—	1	$\bar{B}D$
(m_1, m_9, m_3, m_{11})	—	0	—	1	
$(m_3, m_7, m_{11}, m_{15})$	—	—	1	1	CD
$(m_3, m_{11}, m_7, m_{15})$	—	—	1	1	

- PRIME IMPLICANTS (PI) : Prime Implicant is a group of min terms which cannot be combined with any other min terms or group.
- ESSENTIAL PRIME IMPLICANTS (EPI) : Essential Prime Implicant is a prime implicant in which one or more min terms are unique i.e. it contains atleast one min term, which is not contained in any other PI.

(IV)

Decimal No. Corresponding to min terms	PI	0 1 3 7 8 9 11 15
0, 1, 8, 9	$\overline{B}\overline{C}$	(x) x (x) x
1, 3, 9, 11	$\overline{B}D$	x x x x
3, 7, 11, 15	CD	x (x) x (x)

So, $EPI \rightarrow \overline{B}\overline{C}, CD$

$$\therefore \boxed{Y = \overline{B}\overline{C} + CD}$$

(ii) $Y = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$

(I)	A	B	C	D	
m_0	0	0	0	0	(Zero 1's)
m_1	0	0	0	1	
m_2	0	0	1	0	one 1's
m_8	1	0	0	0	
m_3	0	0	1	1	
m_5	0	1	0	1	(two 1's)
m_9	1	0	0	1	
m_7	0	1	1	1	
m_{11}	1	0	1	1	(three 1's)
m_{14}	1	1	1	0	(ABCD)

(PI)

ABCD is a P.I.

(II)	A	B	C	D
(m_0, m_1)	0	0	0	—
(m_0, m_2)	0	0	—	0
(m_0, m_8)	—	0	0	0
(m_1, m_3)	0	0	—	1
(m_1, m_5)	0	—	0	1
(m_1, m_9)	—	0	0	1
(m_2, m_3)	0	0	1	—
(m_8, m_9)	1	0	0	—
(m_3, m_7)	0	—	1	1
(m_3, m_{11})	—	0	1	1
(m_5, m_7)	0	1	—	1
(m_9, m_{11})	1	0	—	1

(III)	A	B	C	D	
(m_0, m_1, m_2, m_3)	0	0	—	—	} $\rightarrow \bar{A}\bar{B}$
(m_0, m_1, m_8, m_9)	—	0	0	—	
(m_0, m_2, m_1, m_3)	0	0	—	—	} $\rightarrow \bar{B}\bar{C}$
(m_0, m_8, m_1, m_9)	—	0	0	—	
(m_1, m_3, m_5, m_7)	0	—	—	1	} $\rightarrow \bar{A}D$
(m_1, m_3, m_9, m_{11})	—	0	—	1	
(m_1, m_5, m_3, m_7)	0	—	—	1	} $\rightarrow \bar{B}D$
(m_1, m_9, m_3, m_{11})	—	0	—	1	

Seq. P.I $\rightarrow$ ABCD, $\bar{A}\bar{B}$, $\bar{B}\bar{C}$, $\bar{A}D$, $\bar{B}D$

(IV)

Decimal no. Corresponding to min terms.	P.I	0 1 2 3 5 7 8 9 11 14
1) 0, 1, 2, 3	$\bar{A}\bar{B}$	x x (x) x
2) 0, 1, 8, 9	$\bar{B}\bar{C}$	x x (x) x
3) 1, 3, 5, 7	$\bar{A}D$	x x (x) (x)
4) 1, 3, 9, 11	$\bar{B}D$	x x x (x)
5) 14	$ABCD$	(x)

So, EPI $\rightarrow \bar{A}\bar{B}, \bar{B}\bar{C}, \bar{A}D, \bar{B}D, ABCD$

$$\therefore Y = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}D + \bar{B}D + ABCD$$

(iii) $Y = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$

(I)	A B C D	(II)	A B C D
m_1	0 0 0 1	(m_1, m_5)	0 — 0 1 ($\bar{A}\bar{C}D$)(PI)
m_2	0 0 1 0 one 1's	(m_2, m_6)	0 — 1 0
m_4	0 1 0 0	(m_4, m_5)	0 1 0 —
m_5	0 1 0 1	(m_4, m_6)	0 1 — 0
m_6	0 1 1 0 two 1's	(m_4, m_{12})	— 1 0 0
m_{12}	1 1 0 0	(m_5, m_{13})	— 1 0 1
m_{13}	1 1 0 1	(m_6, m_{14})	— 1 1 0
m_{14}	1 1 1 0 three 1's	(m_{12}, m_{13})	1 1 0 —
		(m_{12}, m_{14})	1 1 — 0

(III)	A	B	C	D	
$(m_4, m_5, m_{12}, m_{13})$	—	1	0	—	} $\rightarrow \overline{BC}$
$(m_4, m_6, m_{12}, m_{14})$	—	1	—	0	
$(m_4, m_{12}, m_5, m_{13})$	—	1	0	—	
$(m_4, m_{12}, m_6, m_{14})$	—	1	—	0	

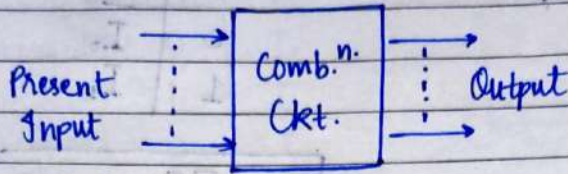
So, PI $\rightarrow \overline{ACD}, \overline{BC}, \overline{BD}$

Decimal no. corresponding to min terms.	P.I	1	5	6	12	13	14
4, 5, 12, 13	$\overline{BC}$	x			x	x	
4, 6, 12, 14	$\overline{BD}$			x	x		x
1, 5	$\overline{ACD}$	x	x				

So, EPI $\rightarrow \overline{BC}, \overline{BD}, \overline{ACD}$

$$\therefore Y = \overline{BC} + \overline{BD} + \overline{ACD}$$

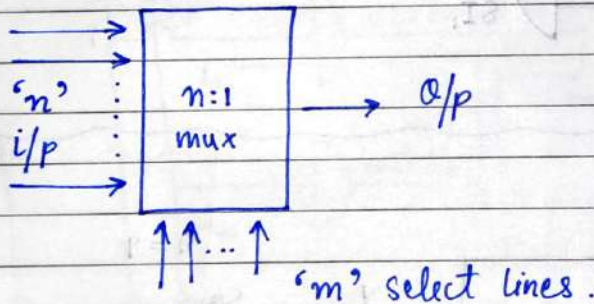
COMBINATIONAL CIRCUIT



- Adder & Subtractor
- Multiplexer
- De-Multiplexer
- Encoder
- Decoder

Multiplexer

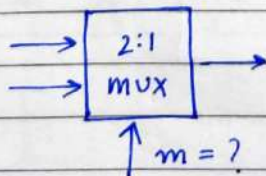
(mux → short form of multiplexer.)



where,

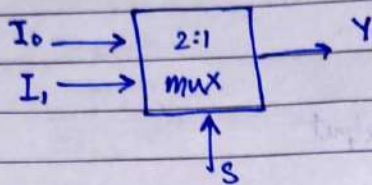
$$n = 2^m$$

for ex : find no. of select lines in the foll. mux :



$$\begin{aligned} n &= 2 \\ \text{So, } n &= 2^m \\ \Rightarrow 2 &= 2^m \\ \Rightarrow m &= 1 \end{aligned}$$

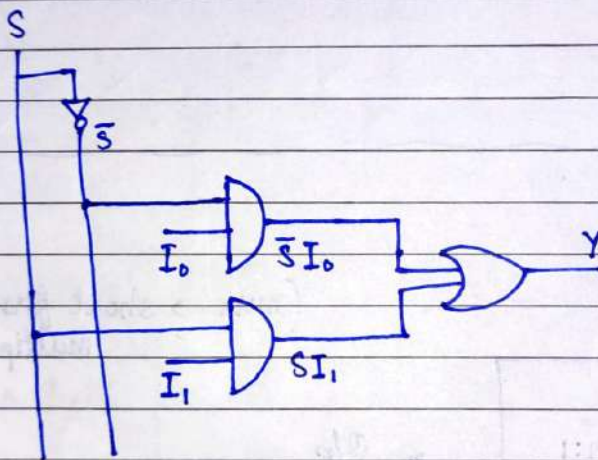
• Truth Table (for 2:1 mux)



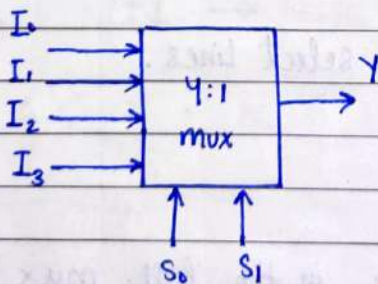
S	Y
0	I_0
1	I_1

$$Y = \bar{S}I_0 + SI_1$$

• Logic Circuit (2:1 mux)



4:1 MUX



$n = 4$

So,

$$n = 2^m$$

$$\Rightarrow 4 = 2^m$$

$$\Rightarrow m = 2$$

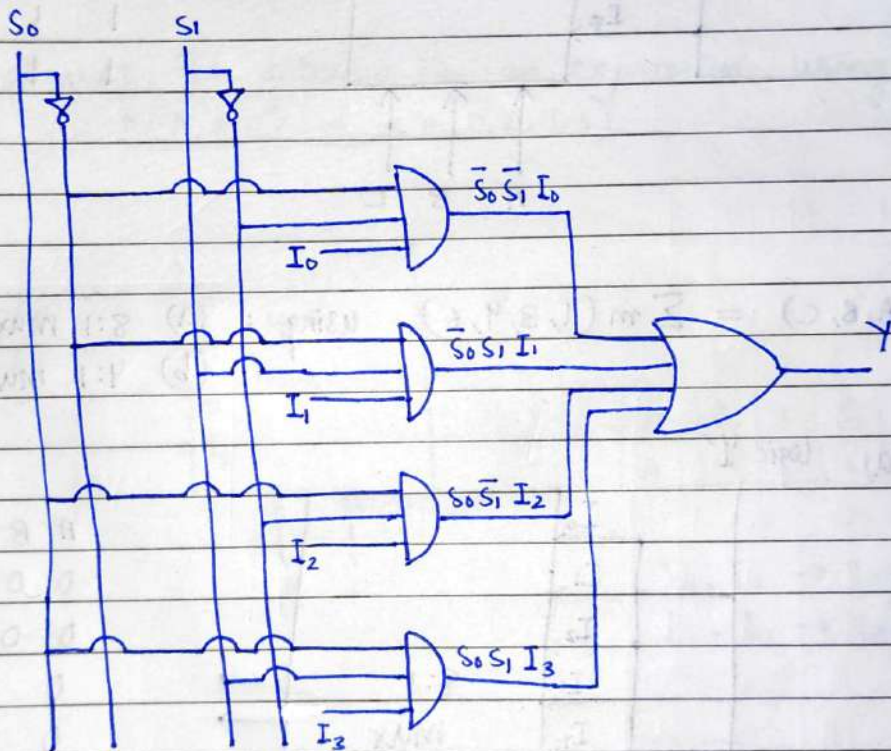
(2 select lines)

• Truth Table (for 4:1 mux)

S_0	S_1	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

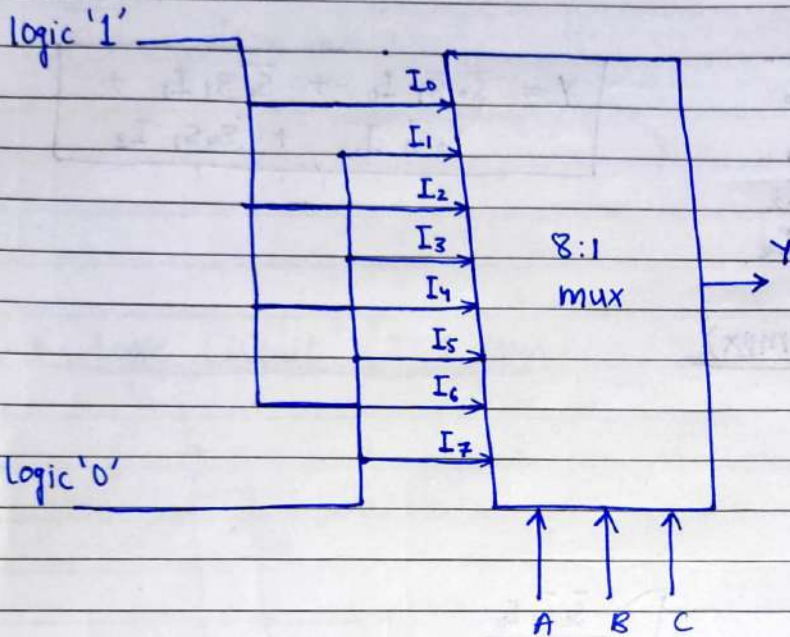
$$Y = \bar{S}_0 \bar{S}_1 I_0 + \bar{S}_0 S_1 I_1 + S_0 \bar{S}_1 I_2 + S_0 S_1 I_3$$

• Logic Circuit (4:1 mux)



Implementation of Logic Expression using mux

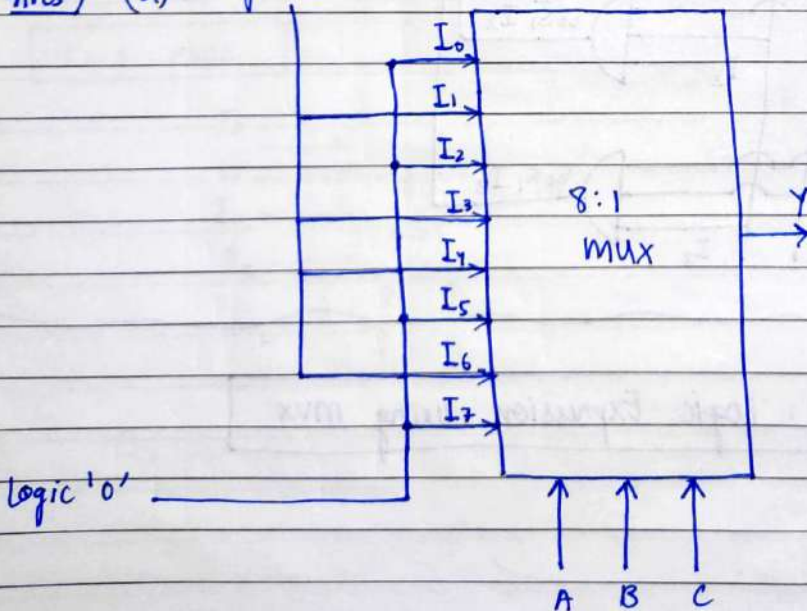
(i) $F(A, B, C) = \sum m(0, 2, 4, 6)$



A	B	C	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

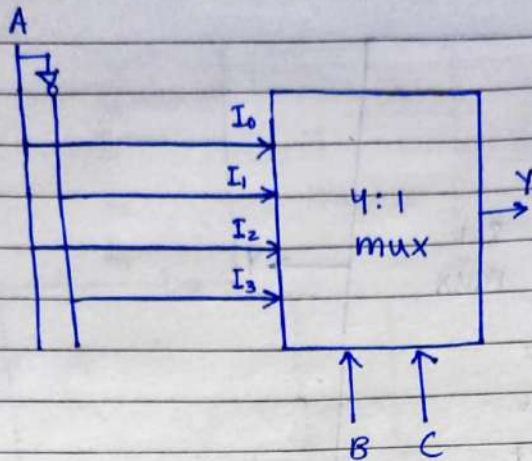
(ii) $F(A, B, C) = \sum m(1, 3, 4, 6)$ using : (a) 8:1 mux.
(b) 4:1 mux.

Ans.) (a) logic '1'



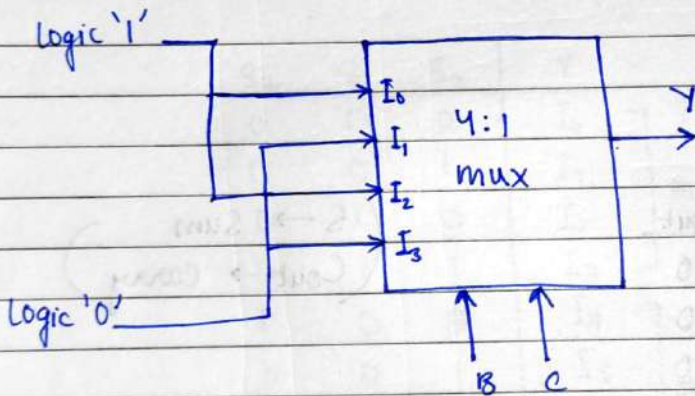
A	B	C	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

(b)



	I_0	I_1	I_2	I_3
$\bar{A}$	0	①	2	③
A	④	5	⑥	7

Ques.) Implement the following Boolean expression using 4:1 mux
 $F(A, B, C) = \sum m(0, 2, 4, 6)$



	I_0	I_1	I_2	I_3
$\bar{A}$	①	1	②	3
A	④	5	⑥	7

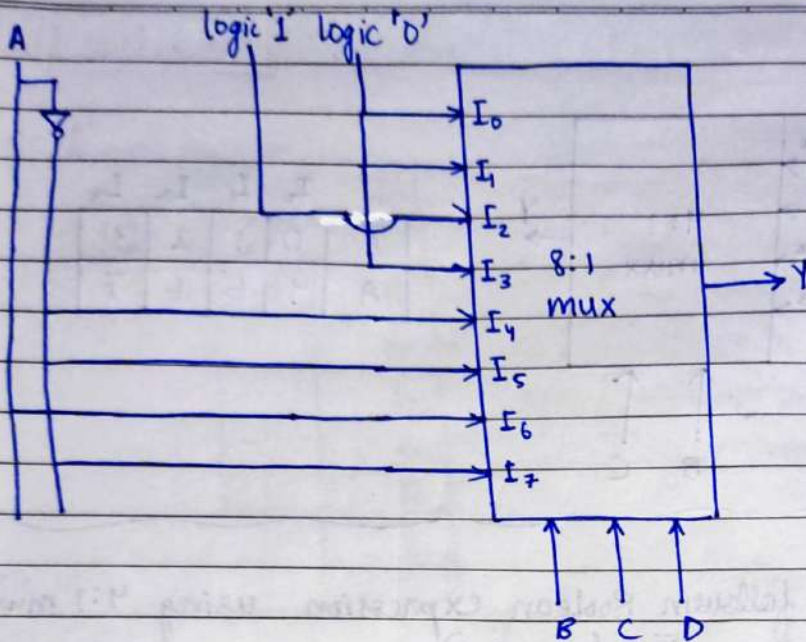
So,

$I_0, I_2 \rightarrow \text{logic '1'}$
 $I_1, I_3 \rightarrow \text{logic '0'}$

Ques.) Implement $F(A, B, C, D) = \sum m(2, 4, 5, 7, 10, 14)$ using 8:1 mux.

Ans.)

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{A}$	0	1	②	3	④	⑤	6	⑦
A	8	9	⑩	11	12	13	⑭	15



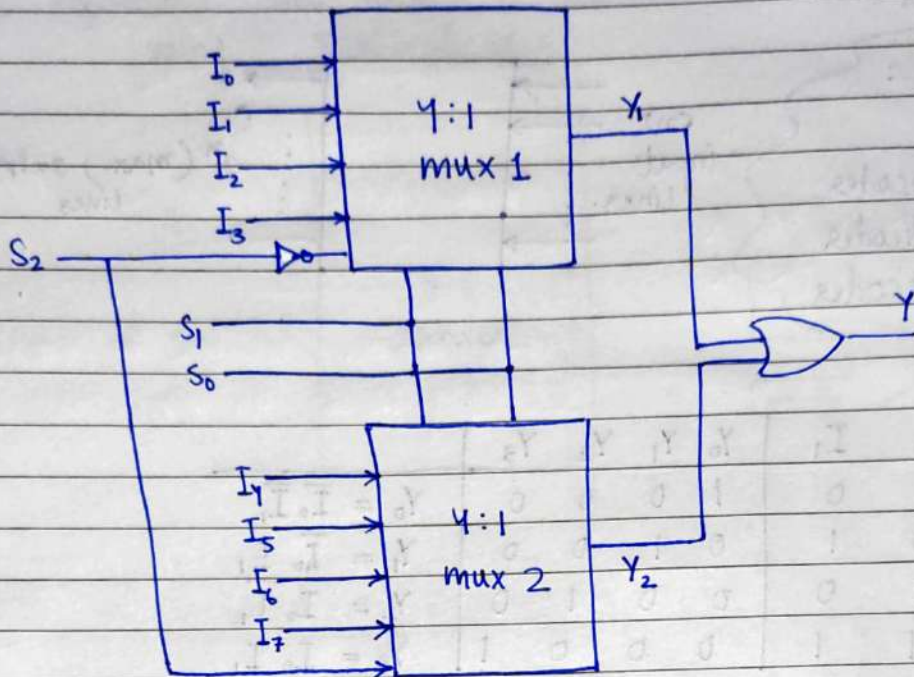
FULL ADDER

Truth Table

A	B	C	S	Count
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

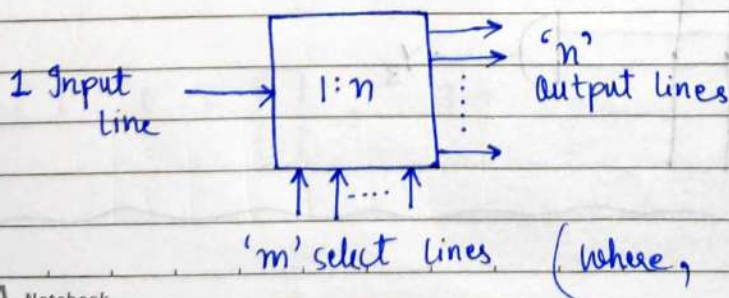
(S → Sum)
(Count → carry)

Implementing FULL ADDER (by obtaining 8:1 mux (using 4:1 mux))



S_2	S_1	S_0	Y	
0	0	0	I_0	MUX 1 enable
0	0	1	I_1	
0	1	0	I_2	
0	1	1	I_3	
1	0	0	I_4	MUX 2 enable.
1	0	1	I_5	
1	1	0	I_6	
1	1	1	I_7	

DE - MULTIPLEXER



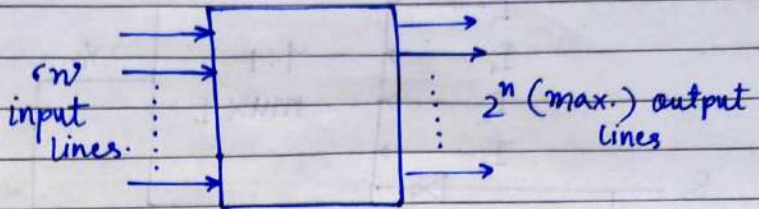
DECODER

We have :

2 : 4 decoder

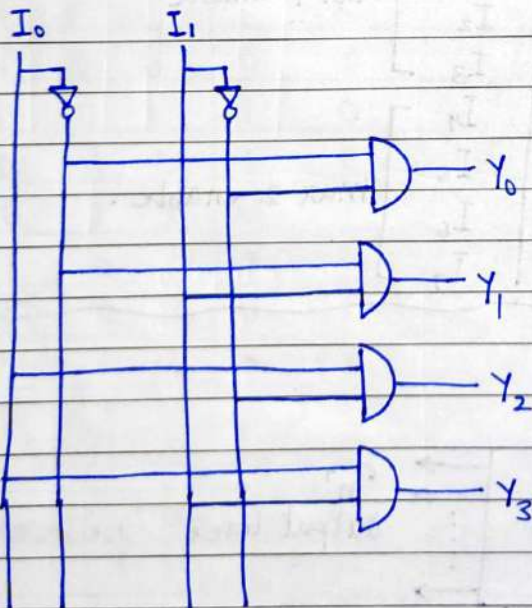
3 : 8 decoder

4 : 16 decoder

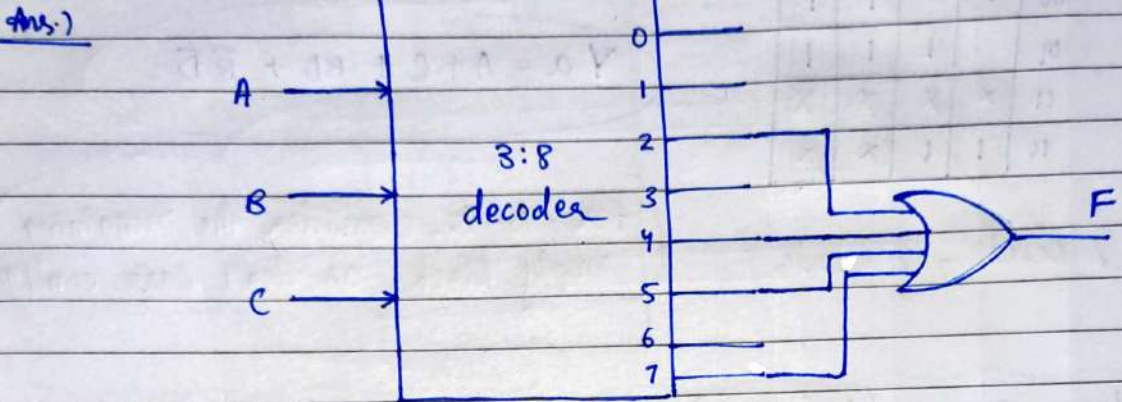


I_0	I_1	Y_0	Y_1	Y_2	Y_3	
0	0	1	0	0	0	$Y_0 = \overline{I_0} \overline{I_1}$
0	1	0	1	0	0	$Y_1 = \overline{I_0} I_1$
1	0	0	0	1	0	$Y_2 = I_0 \overline{I_1}$
1	1	0	0	0	1	$Y_3 = I_0 I_1$

Logic Circuit :

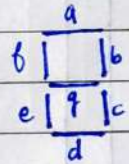


Ques.) Implement the boolean function : $F(A,B,C) = \sum m(2,4,5,7)$ using 3:8 decoder.



BCD to 7 SEGMENT DECODER

	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1



for ex : 0 → , 3 → , 6 → , 7 →

K-Map (for 'a')

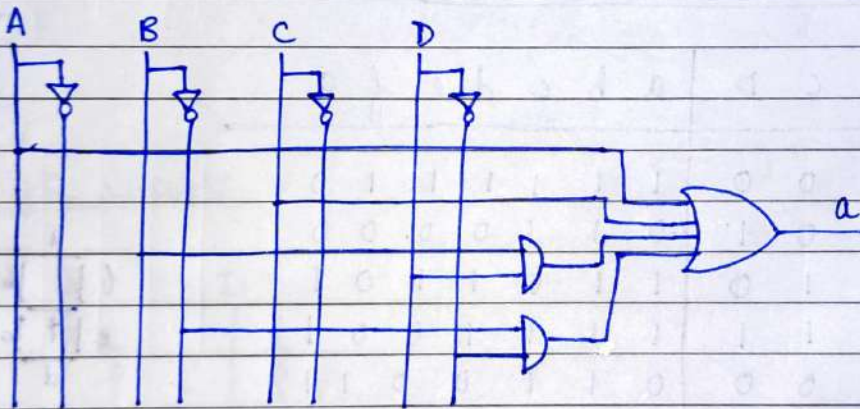
AB \ CD	00	01	11	10
00	1		1	1
01		1	1	1
11	x	x	x	x
10	1	1	x	x

$$a = A + C + BD + \bar{B}\bar{D}$$

(2 octets, 2 Quads)

(for 10-15, consider the remaining empty blocks as don't care condⁿ)

Logic Circuit (for 'a')

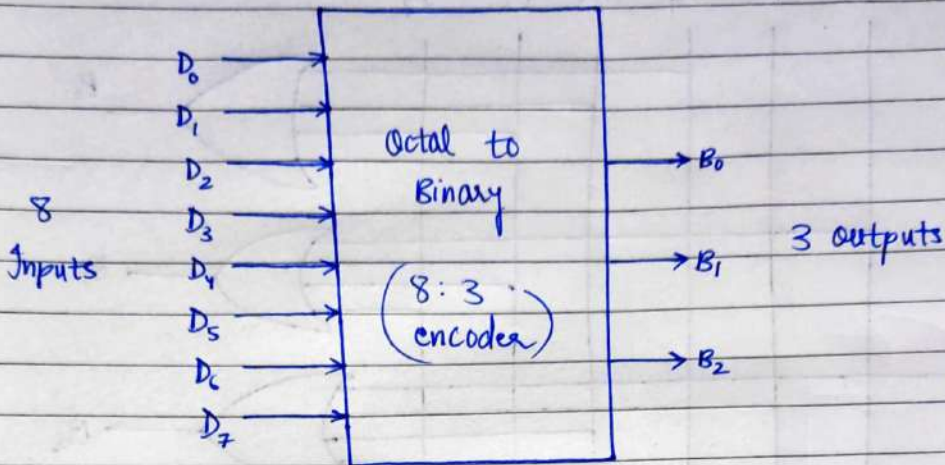


ENCODER

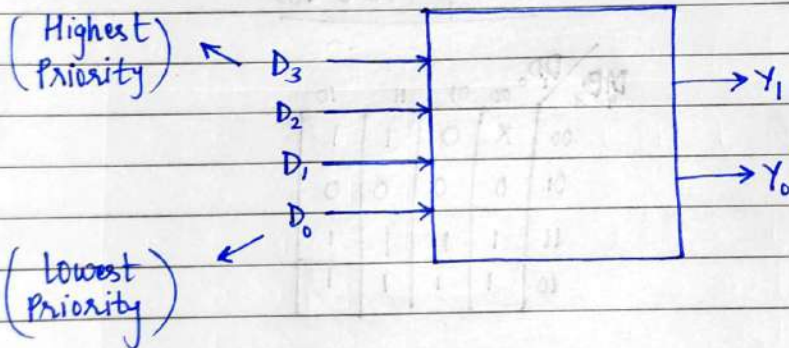
(Inputs)

(Outputs)

D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	B ₀	B ₁	B ₂
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1



PRIORITY ENCODER

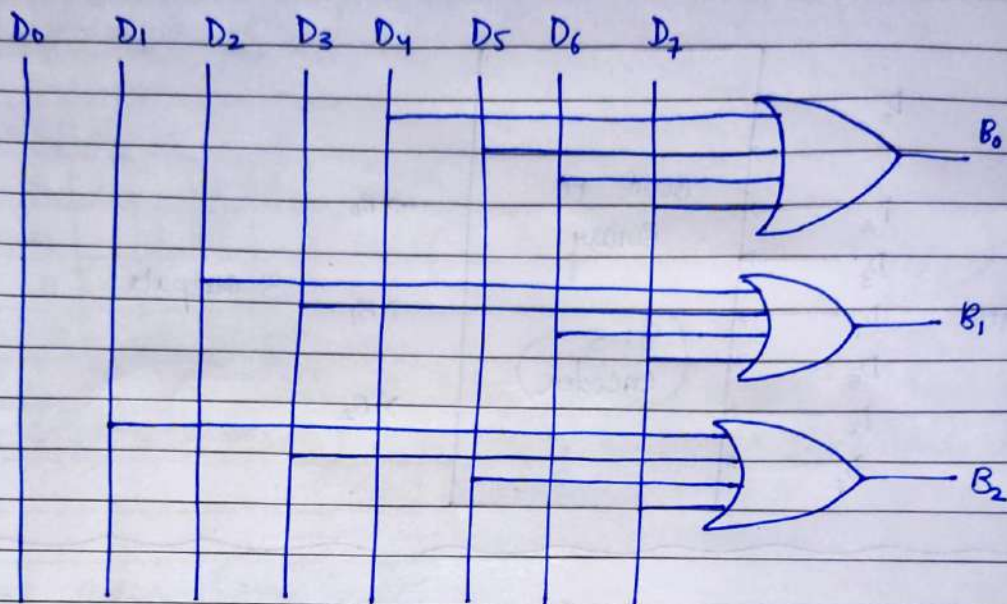


D ₃	D ₂	D ₁	D ₀	Y ₁	Y ₀
0	0	0	0	X	X
0	0	0	1	0	0
0	0	1	X	0	1
0	1	X	X	1	0
1	X	X	X	1	1

$$B_0 = D_4 + D_5 + D_6 + D_7$$

$$B_1 = D_2 + D_3 + D_6 + D_7$$

$$B_2 = D_1 + D_3 + D_5 + D_7$$



• K-Map (for Y_1)

$D_3 \backslash D_0$	00	01	11	10
00	X	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$Y_1 = D_2 + D_3$$

• K-Map (for Y_0)

$D_3 \backslash D_0$	00	01	11	10
00	X	0	1	1
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$Y_0 = D_3 + \bar{D}_2 D_1$$

• Circuit Diagram

